

Guided Notes: The Logistic-Regression Negative Log-Likelihood

Name: _____

Partner(s): _____

Date: _____

Learning goals

By the end of these notes, you should be able to

1. define the linear predictor, sigmoid probability, and Bernoulli negative log-likelihood;
 2. use the chain rule to show why the logistic-loss derivative simplifies to $p_i - y_i$;
 3. derive the gradients with respect to the weight vector w and intercept b ; and
 4. interpret the gradient and write it in vectorized matrix form.
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1. Setup and notation

We observe n labeled examples. For observation i , x_i is a feature vector and y_i is a binary label.

Symbol	Meaning
x_i	Feature vector for observation i
y_i	Binary label
w	Weight vector
b	Bias/intercept
z_i	Linear predictor
p_i	Predicted probability that $y_i = 1$

The model first computes a linear score and then converts that score to a probability:

$$z_i = w^\top x_i + b, \quad p_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}.$$

Quick dimension check

Assume $x_i, w \in \mathbb{R}^d$ are column vectors. Complete each dimension.

Expression	Dimensions
$w^\top x_i$	_____
z_i	_____
p_i	_____
Xw , where $X \in \mathbb{R}^{n \times d}$	_____
$b\mathbf{1}$, where $\mathbf{1} \in \mathbb{R}^n$	_____

Think first. Why must $w^\top x_i$ be a scalar before we can add b ?

2. From probability to a loss

For one observation, the Bernoulli probability model can be written compactly as

$$P(Y_i = y_i \mid x_i) = p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Derive the single-observation log-likelihood

Take the natural logarithm of the Bernoulli probability above. Use $\log(a^c) = c \log(a)$.

$$\ell_i(w, b) = \log P(Y_i = y_i \mid x_i)$$

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Derive the full-data log-likelihood and loss

Assuming the observations are conditionally independent, add the single-observation log-likelihoods:

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1. If $y_i = 1$, which term remains in the log-likelihood? What happens when p_i is close to 1 or close to 0?
2. If $y_i = 0$, which term remains?
3. Why do we minimize the **negative** log-likelihood rather than maximize the log-likelihood?

4. What is the difference between $J(w, b)$ and the mean negative log-likelihood $J(w, b)/n$? Does it change the minimizing parameter values?

3. The sigmoid derivative

The sigmoid is

$$\sigma(z) = \frac{1}{1 + e^{-z}} = (1 + e^{-z})^{-1}.$$

Derive the derivative

Differentiate using the chain rule. Keep the derivative of the inner expression visible on its own line.

$$\frac{d\sigma(z)}{dz} = \frac{d}{dz}(1 + e^{-z})^{-1}$$

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Rewrite your final expression in terms of $\sigma(z)$ and $1 - \sigma(z)$:

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad 1 - \sigma(z) = \frac{\quad}{1 + e^{-z}}.$$

Use those identities to complete the derivative:

$$\frac{d\sigma(z)}{dz} =$$

For observation i ,

$$\frac{dp_i}{dz_i} =$$

4. Guided derivation for one observation

We now derive the derivative of J_i with respect to the linear predictor z_i . This is the key step: once it is known, the gradients with respect to w and b follow from the chain rule.

Recall

$$J_i = -\left[y_i \log p_i + (1 - y_i) \log(1 - p_i)\right].$$

Step 1: Differentiate with respect to p_i

Treat y_i as a constant. Work out the two component derivatives first, including the inner derivative of $1 - p_i$.

$$\frac{d}{dp_i} (y_i \log p_i) =$$

$$\frac{d}{dp_i} ((1 - y_i) \log(1 - p_i)) =$$

=

Now combine those results with the leading negative sign in J_i :

$$\frac{dJ_i}{dp_i} =$$

Step 2: Apply the chain rule

Use

$$\frac{dJ_i}{dz_i} = \frac{dJ_i}{dp_i} \frac{dp_i}{dz_i}.$$

Substitute your derivative from Step 1 and your sigmoid derivative from Part 3. Cancel denominators and show the algebra:

$$\frac{dJ_i}{dz_i} =$$

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The central result is

$\frac{dJ_i}{dz_i} =$

Interpret the result

1. If $p_i > y_i$, what sign does the derivative have?
2. If $y_i = 1$ and the model predicts p_i close to 0, is the error signal $p_i - y_i$ large or small? What does that suggest about the update?
3. Why is $p_i - y_i$ called a residual in probability space?

5. Gradient with respect to the weights

The linear predictor is

$$z_i = w^\top x_i + b.$$

Using the column-gradient convention, complete

$$\frac{\partial z_i}{\partial w} = \quad .$$

Your turn: record the full weight gradient

$\nabla_w J(w, b) =$

What does each observation contribute?

The observation-level contribution is

Explain the role of each factor:

Factor	Interpretation
$p_i - y_i$	
x_i	
Your observation-level contribution	

6. Gradient with respect to the bias

Because b appears in z_i with coefficient 1,

$$\frac{\partial z_i}{\partial b} = \quad .$$

Your turn: record the full bias gradient

$\frac{\partial J(w, b)}{\partial b} =$

Compare the two gradients. How does the weight gradient differ from the bias gradient?

7. Vectorized form

Let $X \in \mathbb{R}^{n \times d}$ have row i equal to $x_i^\top$, and let $y, p \in \mathbb{R}^n$ collect the labels and predicted probabilities.

$$z = Xw + b\mathbf{1}, \quad p = \sigma(z),$$

where σ is applied element by element.

Define the residual vector

$$r = p - y.$$

Use the observation-level formulas to fill in the matrix expressions:

$$\nabla_w J =$$

$$\frac{\partial J}{\partial b} =$$

Dimension check

Expression	Dimensions
$p - y$	_____
$X^\top(p - y)$	_____
$\mathbf{1}^\top(p - y)$	_____

If the mean negative log-likelihood is used instead,

$$\bar{J} = \frac{1}{n} J,$$

then both gradients are multiplied by

.

8. Small numerical check

Consider two observations with one feature:

i	x_i	y_i
1	1	1
2	2	0

Start at $w = 0$ and $b = 0$.

Part A: Compute probabilities and residuals

Complete the table.

i	$z_i = wx_i + b$	$p_i = \sigma(z_i)$	$p_i - y_i$
1	_____	_____	_____
2	_____	_____	_____

Part B: Compute both gradient components

$$\frac{\partial J}{\partial b} = \sum_{i=1}^2 (p_i - y_i) =$$

$$\frac{\partial J}{\partial w} = \sum_{i=1}^2 (p_i - y_i)x_i =$$

Part C: Interpret the direction

Suppose we take one gradient-descent step with learning rate $\alpha > 0$:

$$w_{\text{new}} = w - \alpha \frac{\partial J}{\partial w}, \quad b_{\text{new}} = b - \alpha \frac{\partial J}{\partial b}.$$

1. Does w increase or decrease? Explain using the sign of its gradient.

2. Does b change for this particular starting point and dataset? Why?

9. Wrap-up: the gradient recipe

For each observation, follow this chain:

$$x_i \longrightarrow z_i = w^\top x_i + b \longrightarrow p_i = \sigma(z_i) \longrightarrow J_i \longrightarrow \frac{dJ_i}{dz_i} = p_i - y_i.$$

Then combine the observation-level error signal with the input:

$$\boxed{\nabla_w J = \sum_{i=1}^n (p_i - y_i) x_i, \quad \frac{\partial J}{\partial b} = \sum_{i=1}^n (p_i - y_i).}$$

Before you finish, explain in your own words why the sigmoid derivative and the log-loss derivative cancel so cleanly.

Common checks

- Did you distinguish z_i , p_i , and y_i ?
- Did you preserve the negative sign in the negative log-likelihood?
- Did you apply the chain rule through $p_i = \sigma(z_i)$?
- Does $\nabla_w J$ have the same dimension as w ?
- Did you distinguish a sum loss from a mean loss?